

Optimal Evasion in an Active Target Defense Scenario

Isaac E. Weintraub^{*}, Eloy Garcia[†], and David Casbeer[‡]

Control Science Center of Excellence, Air Force Research Laboratory, Wright-Patterson AFB, OH 45433

Meir Pachter[§]

Air Force Institute of Technology, Wright-Patterson AFB, OH 45433

A three-agent scenario is considered in 3-D Cartesian space, where a Pursuer aims at intercepting an Evader while the Evader aims to out-maneuver the Pursuer. A Defender working in cooperation with the Evader aims at intercepting the Pursuer before the Pursuer can capture the Evader. A proportional navigation strategy is implemented by the Defender and the Pursuer. The optimal control problem which strives at solving for the heading rate and flight path angle rate of the Evader is posed and solved. Using the even collocation method, the optimal controls for the Evader are found using a nonlinear program solver. An example is provided to illustrate the maneuvers made by the Evader to escape the Pursuer with maximum separation at final time.

Nomenclature

g	=	Inequality constraint
h	=	Equality constraint
$\mathbf{h}$	=	Equality constraint vector
$\mathcal{H}$	=	Hamiltonian
J	=	Objective Cost Functional
k	=	Discrete time step index
$\mathbf{M}$	=	Matrix for costate dynamics
N	=	Total number of discrete time step
p	=	Co-states
R_{xy}	=	Line of sight from an agent x to agent y
t	=	Time
$\mathbf{u}$	=	Input vector
$\dot{\psi}_E(t)$	=	Heading rate command
$\dot{\gamma}_E$	=	Flight path angle rate command
V	=	Velocity of an agent
x	=	X position of an agent
$\mathbf{x}$	=	States
y	=	Y position of an agent
z	=	Z position of an agent
α	=	Longitudinal line of sight angle
β	=	Lateral line of sight angle
γ	=	Flight path angle of an agent
σ	=	Pursuit Angle (angle between an agent's velocity vector and the line of sight to his target)
ϕ	=	Escape Angle (angle between the agent velocity vector and line of sight from its pursuer)
ξ	=	Arbitrary variable used for compactness
ψ	=	Heading of an agent
ω	=	Angular rate of motion

Subscript

^{*}Control Science Center of Excellence, Air Force Research Laboratory, Wright-Patterson AFB, OH 45433, AIAA Senior Member

[†]Control Science Center of Excellence, Air Force Research Laboratory, Wright-Patterson AFB, OH 45433, AIAA Senior Member

[‡]Control Science Center of Excellence, Air Force Research Laboratory, Wright-Patterson AFB, OH 45433, AIAA Associate Fellow

[§]Department of Electrical Engineering, Air Force Institute of Technology, Wright-Patterson AFB, OH 45433, AIAA Associate Fellow

D	=	Defender
E	=	Evader
f	=	Final condition (final state)
LAT	=	Lateral component
LON	=	Longitudinal component
P	=	Pursuer
0	=	Zero condition (initial state)
<i>Superscript / Overscript</i>		
$*$	=	Optimal
$\cdot$	=	Time rate of change

I. Introduction

The defense of airborne assets is an area of interest for the military and is commonly referred to as defensive counter air (DCA) operations. In an effort to increase an aircraft's survivability against incoming threats, it is important to pose and solve general engagement scenarios in order to ascertain useful strategies and understand the particular objectives of pursuit-evasion. The concept and aims of DCA has been outlined in a joint-publication [1]. In the Joint publication, the importance of countermeasures such as decoys, chaffs, flares, electronic counter measures, and evasive maneuvers have been described. In this paper, the only defensive strategy being considered is that of evasive maneuvers made by the evading target, called the Evader. Other defensive tactics and considerations are considered out of the scope of this work.

The concept of posing and solving optimal strategies for multi-agent pursuit-evasion scenarios was considered in a seminal work by Isaacs [2]. In his treatise, Isaacs considered pursuit-evasion as a differential game wherein two players, in opposition with each other, aim at either maximizing or minimizing a common objective cost functional. The agent who aims at minimizing the objective cost functional is considered the Pursuer, while the agent which strives at maximizing the same objective cost functional is called the Evader. The motion or dynamics of each player was governed by a set of differential equations, forming a differential game. In the event that the strategy of all but one of the agents is pre-determined, the differential game becomes a single-sided optimization problem, more commonly referred to as an optimal control problem.

The goal of optimal control theory is to find the commands which drive a system in a manner which minimize or maximize a given objective subject to system constraints, control limitations, boundary conditions, and path constraints [3, 4]. Two popular method of solving optimal control methods are the so-called indirect and direct methods. The indirect methods make use of the calculus of variations [5] whereby a Hamiltonian is formed and set of costates are used to formulate necessary optimality conditions. By solving the set of optimality conditions, candidate optimal controls are found which solve the optimal control problem. In the event of control limitations, Pontryagin's Minimum Principle (PMP) is used [6]. Alternatively, one may solve optimal control problems directly, by use of a nonlinear program solver. One popular method of posing and solving optimal control problems directly is through the use of collocated optimization methods [7]. These collocated optimization methods may be used to solve optimal control problems and are less sensitive to initial guess than the popular shooting method [8].

Active target defense (ATD) has been considered in the past [9–15]. Boyell considered the active defense against an incoming Torpedo closing on a target ship. In order to protect itself, the ship launches a counter-weapon aimed at intercepting the incoming threat and consequently, providing the necessary defense [9, 10]. Fuchs, Kahrgonekar, and Evers considered the three-agent differential game consisting of a single pursuer and two evaders. In the differential game, optimal evader and pursuer strategies were found and presented [11] which guaranteed either successful capture or escape of the Evaders from the Pursuer. Rantoo and Shima considered a three-agent aerial engagement, similar to that of Boyell [9]. In their paper [12], a target aircraft launches a defensive missile to counter the incoming pursuing missile. The incoming threat was considered to be guided by either proportional navigation or pure pursuit. Venkatesan and Sinha proposed a application of an optimal guidance law derived from the solution of a differential game involving the defense of a static target from a pursuer [13]. Using the optimal solution, a heuristic application of the optimal control was applied to a maneuvering target. Later, the solution to the differential game was presented in 2-D [14] and in 3-D [15]. The team involving Garcia, Casbeer, and Pachter have investigated the active target defense scenario in great detail [16–20]. In one of their papers, the considered the Pursuer to use either proportional navigation or pure pursuit as a guidance strategy, similar to [12]. The team composed of the Defender and Evader share their strategies and information in the differential game formulation. The team involving Garcia, Casebeer, and Pachter, also considered the scenario where the Evader launches a Defending asset at the initial time [17]. The active target defense differential game

was investigated in their work [18]; the solution of the ATDDG provided the instantaneous optimal heading angles for each agent playing the game in order to achieve each of their objectives. A thorough investigation of the ATDDG was presented in [19]. In the paper, both a presentation of the game of kind and game of degree for the ATDDG was presented. Many significant papers have investigated active target defense in the 2-D plane; however, less common are active target defense problems in 3-D Cartesian space, which is the aim of this work.

Some related work to this problem in 3-D have been investigated in the past. In an early work by Shinar and Gutman, a missile-aircraft scenario was considered in 3-D space [21]. Their paper presented the avoidance of a known pursuer by an evader who had no state information about the evading aircraft. Shinar also found and reported solution techniques for realistic pursuit-evasion games involving missile and aircraft [22]. Over a decade later, Greenwood considered an aerial dogfight scenario in 3-D [23]. In his paper, Greenwood considered more complex models, considering thrust, drag, and lift of both aircraft. Greenwood performed his analysis using a spherical polar-coordinate system to describe the system involving both aircraft. Later, Ehtamo and Raivio posed and solved the Pursuit-Evasion differential game between two aerial adversaries, taking into account boundaries in each player's envelope [24]. Imado and Kuroda considered a discrete approach to solving the missile-aircraft pursuit-evasion differential game rather than consider the engagement dynamics outright [25]. In a closely related work, Garcia et al. posed and solved the active target defense differential game (ATDDG) where an attacker missile tries to capture a target aircraft which is significantly slower than the target aircraft [26]. In their setup the defender was provided with the target aircraft's information and was endowed with a capture radius. In this paper, the Defender considered point capture and is not provided state information about the Evader. In a work by Saurav, Kumar, and Maity, a finite-time horizon optimal controller is designed for the three-agent active target defense scenario in 3-D [27]. In their paper, they provide target information to the defender and consider more complete equations of motion for the missiles including gravity, thrust, and drag. Finally, in a recent work, Garcia, Casbeer, and Pachter, posed a differential game in 3-D wherein two pursuers form a team to guard a single target against an evader with same speed as the pursuers [28].

In this paper, a three-agent scenario is considered. In the scenario, an Evader-Defender team aims at preventing a faster Pursuer from capturing the Evader. The Defender and Pursuer strategies are assumed to be proportional navigation as described in [29] and [30, pg119]. Proportional navigation accomplishes capture of an evader by providing commands which zero the line of sight rate, ensuring capture [31]. Because the Defender and the Pursuer are using proportional navigation, the Defender does not have any information about the Evader and as a result, the Evader must maneuver in a manner which aids the Defender in his capture of the Pursuer before the Pursuer can Capture the Evader. This scenario was considered in a previous work in 2-D [32], and this paper aims at investigating the strategy of the Evader when the problem is extended to 3-D.

This paper is organized as follows: In Section II the problem is formulated and equations of motion are presented in both Cartesian and relative polar spaces. Next, Section III sets up the optimal control problem, providing the necessary conditions for optimality, boundary conditions, and control limits. In Section IV, the equal collocation method used to solve the optimal control problem is presented. Next an example is setup and solved in Section V. Finally, concluding remarks are made in Section VI.

II. Model

The Pursuer-Evader-Defender engagement scenario in three-dimensional Cartesian space is illustrated in Figure 1. It is assumed that the velocities of the Pursuer, Evader, and Defender are assumed constant; these velocities are denoted respectively by v_P , v_E , and v_D . Using Cartesian coordinates, the simple dynamics of the three agents are as follows:

$$\begin{aligned} \dot{x}_P &= v_P \cos \gamma_P \cos \psi_P & \dot{x}_E &= v_E \cos \gamma_E \cos \psi_E & \dot{x}_D &= v_D \cos \gamma_D \cos \psi_D \\ \dot{y}_P &= v_P \cos \gamma_P \sin \psi_P & \dot{y}_E &= v_E \cos \gamma_E \sin \psi_E & \dot{y}_D &= v_D \cos \gamma_D \sin \psi_D \\ \dot{z}_P &= v_P \sin \gamma_P & \dot{z}_E &= v_E \sin \gamma_E & \dot{z}_D &= v_D \sin \gamma_D \end{aligned} \quad (1)$$

The input to the system is the heading rate and flight path angle rate of the Evader:

$$\mathbf{u}(t) = \{\dot{\psi}_E, \dot{\gamma}_E\} \quad (2)$$

Using this formulation, ψ_x represents the heading angle and γ_x represents the flight path angle of an agent x . In general (as well as this paper), the Pursuer is faster than the Evader, so that $v_P > v_E$. The Defender is assumed to be superior to the Pursuer in speed, *i.e.*, $v_D > v_P$. It should be noted that the Pursuer may be faster than the Defender or equal in speed; however, this is left for future work and analysis.

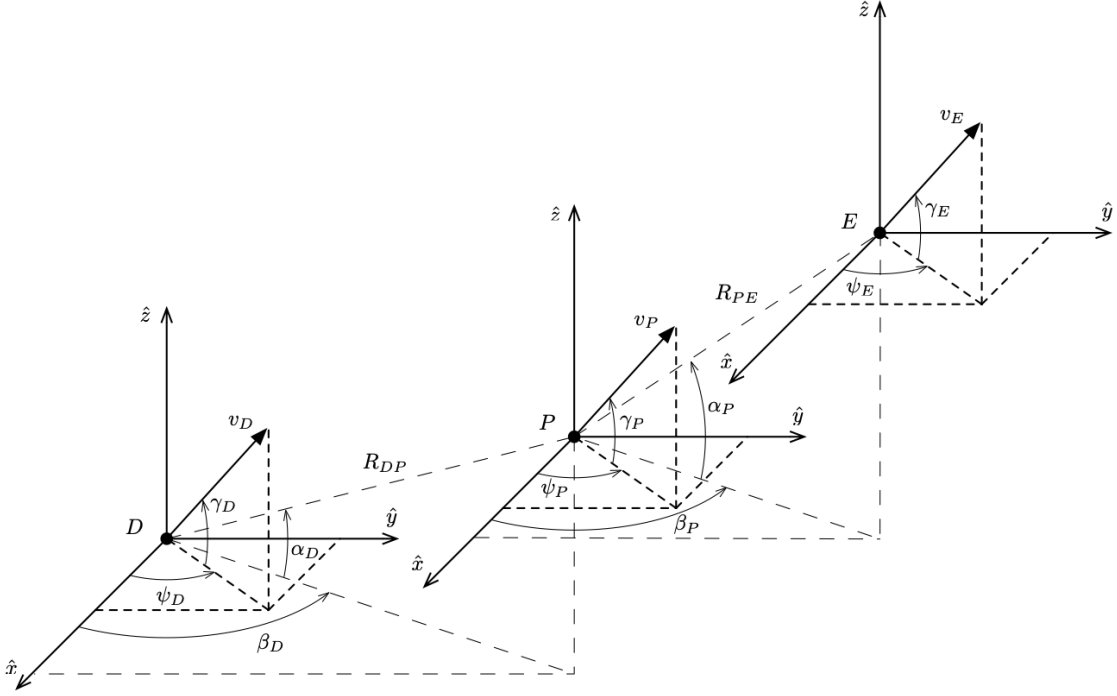


Fig. 1 Pursuer-Evader-Defender Scenario Geometry

Using the Cartesian space, the line of sight distance between agents is defined as the two-norm distance. The line of sight distance between the Pursuer and the Evader is R_{PE} and the line of sight distance between the Defender and the Pursuer is R_{DP} . Further, define the azimuth and elevation of the Evader in the local frame of the Pursuer to be β_P and α_P respectively. Similarly, the azimuth and elevation of the Pursuer in the local frame of the Defender is β_D and α_D respectively. The equations of motion are developed referencing Figure 1. The relative line of sight angles in the lateral and longitudinal planes are found as follows:

$$\begin{aligned}\alpha_D &= \tan^{-1} \left(\frac{z_P - z_D}{\sqrt{(y_P - y_D)^2 + (x_P - x_D)^2}} \right) & \beta_D &= \tan^{-1} \left(\frac{y_P - y_D}{x_P - x_D} \right) \\ \alpha_P &= \tan^{-1} \left(\frac{z_E - z_P}{\sqrt{(y_E - y_P)^2 + (x_E - x_P)^2}} \right) & \beta_P &= \tan^{-1} \left(\frac{y_E - y_P}{x_E - x_P} \right).\end{aligned}\quad (3)$$

Using the line of sight angles α_x and β_x for an agent x , the angular difference between an agent's velocity and the heading to the pursued agent is obtained. For this derivation, the longitudinal and lateral directions have been separated.

$$\begin{aligned}\sigma_{LAT,D} &= \beta_D - \psi_D & \sigma_{LAT,P} &= \beta_P - \psi_P \\ \sigma_{LON,D} &= \alpha_D - \gamma_D & \sigma_{LON,P} &= \alpha_P - \gamma_P\end{aligned}\quad (4)$$

Similarly, taking the difference between an agent's heading and the pursuer's line of sight angles provides the angle ϕ_x . By considering the longitudinal and lateral planes separately, the following is obtained:

$$\begin{aligned}\phi_{LAT,P} &= \beta_D - \psi_P & \phi_{LAT,E} &= \beta_P - \psi_E \\ \phi_{LON,P} &= \alpha_D - \gamma_P & \phi_{LON,E} &= \alpha_P - \gamma_E\end{aligned}\quad (5)$$

Next, the range between the Defender and the Pursuer and the range between the Pursuer and the Evader can be obtained using the two-norm distance:

$$R_{DP} = \sqrt{(x_P - x_D)^2 + (y_P - y_D)^2 + (z_P - z_D)^2} \quad R_{PE} = \sqrt{(x_E - x_P)^2 + (y_E - y_P)^2 + (z_E - z_P)^2} \quad (6)$$

Using the pursuit angles from Equation (4) and the escape angles from Equation (5), the line of sight rates in the lateral and longitudinal directions are obtained just as they were in [32]:

$$\begin{aligned}\dot{\beta}_P &= \frac{-(v_E \sin \phi_{LAT,E} - v_P \sin \sigma_{LAT,P})}{R_{PE}} & \dot{\beta}_D &= \frac{-(v_P \sin \phi_{LAT,P} - v_D \sin \sigma_{LAT,D})}{R_{DP}} \\ \dot{\alpha}_P &= \frac{-(v_E \sin \phi_{LON,E} - v_P \sin \sigma_{LON,P})}{R_{PE}} & \dot{\alpha}_D &= \frac{-(v_P \sin \phi_{LON,P} - v_D \sin \sigma_{LON,D})}{R_{DP}}\end{aligned}\quad (7)$$

The line of sight rates in Equation (7) are used to formulate the proportional navigation controls for the Pursuer and Defender. The heading rate and flight path angle rate for both the Pursuer and Defender are composed of a multiplication of a gain and the line of sight rates. The gain for the Defender and Pursuer are N_D and N_P respectively. The gains for both the Pursuer and Defender in this paper are the same, and the traditional value of $N = 3$ is used for both agents.

The nonlinear dynamics of the pursuit-evasion scenario, which are written in the form $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t))$ as follows:

$$\begin{pmatrix} \dot{x}_P \\ \dot{y}_P \\ \dot{z}_P \\ \dot{\psi}_P \\ \dot{\gamma}_P \\ \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \\ \dot{\psi}_E \\ \dot{\gamma}_E \\ \dot{x}_D \\ \dot{y}_D \\ \dot{z}_D \\ \dot{\psi}_D \\ \dot{\gamma}_D \end{pmatrix} = \begin{pmatrix} v_P \cos \gamma_P(t) \cos \psi_P(t) \\ v_P \cos \gamma_P(t) \sin \psi_P(t) \\ v_P \sin \gamma_P(t) \\ N_P \dot{\beta}_P(t) \\ N_P \dot{\alpha}_P(t) \\ v_E \cos \gamma_E(t) \cos \psi_E(t) \\ v_E \cos \gamma_E(t) \sin \psi_E(t) \\ v_E \sin \gamma_E(t) \\ \dot{\psi}_E(t) \\ \dot{\gamma}_E(t) \\ v_D \cos \gamma_D(t) \cos \psi_D(t) \\ v_D \cos \gamma_D(t) \sin \psi_D(t) \\ v_D \sin \gamma_D(t) \\ N_D \dot{\beta}_D \\ N_D \dot{\alpha}_D \end{pmatrix} \quad (8)$$

Although Equation (8) fully describes a set of ordinary differential equations which represents the problem at hand, it may be reduced into a relative polar frame. This reduction in the number of required states assists with numerical computation of optimal solutions. Once the optimal control is found, one may simply substitute the optimal control back into Equation (8) to obtain the optimal solution in Cartesian space.

The relative polar representation of the dynamics is formed by starting with the velocities along the line of sights and representing the rate of closure as a function of the angles in Equations (4) and (5). Using the Euler rotations about the z -axis with lateral angle then the rotation about the x -axis with the longitudinal angle, the rate change of the distance between two agents may be obtained as follows:

$$\begin{aligned}\dot{R}_{DP} &= -v_D (\cos \sigma_{LAT,D} \cos \sigma_{LON,D} + \cos \sigma_{LAT,D} + \cos \sigma_{LON,D} - 1) / 2 \\ &\quad + v_P (\cos \phi_{LAT,P} \cos \phi_{LON,P} + \cos \phi_{LAT,P} + \cos \phi_{LON,P} - 1) / 2 \\ \dot{R}_{PE} &= -v_P (\cos \sigma_{LAT,P} \cos \sigma_{LON,P} + \cos \sigma_{LAT,P} + \cos \sigma_{LON,P} - 1) / 2 \\ &\quad + v_E (\cos \phi_{LAT,E} \cos \phi_{LON,E} + \cos \phi_{LAT,E} + \cos \phi_{LON,E} - 1) / 2\end{aligned}\quad (9)$$

Continuing the transformation from the Cartesian to relative polar space, the pursuit angle dynamics may be found by differentiating eq. (4) with respect to time and substituting the line of sight dynamics accordingly from Equation (7).

$$\begin{aligned}\dot{\sigma}_{LAT,D} &= \dot{\beta}_D - \dot{\psi}_D = (1 - N_D) \dot{\beta}_D = \frac{(N_D - 1)(v_P \sin \phi_{LAT,P} - v_D \sin \sigma_{LAT,D})}{R_{DP}} \\ \dot{\sigma}_{LON,D} &= \dot{\alpha}_D - \dot{\gamma}_D = (1 - N_D) \dot{\alpha}_D = \frac{(N_D - 1)(v_P \sin \phi_{LON,P} - v_D \sin \sigma_{LON,D})}{R_{DP}} \\ \dot{\sigma}_{LAT,P} &= \dot{\beta}_P - \dot{\psi}_D = (1 - N_P) \dot{\beta}_P = \frac{(N_P - 1)(v_E \sin \phi_{LAT,E} - v_P \sin \sigma_{LAT,P})}{R_{PE}} \\ \dot{\sigma}_{LON,P} &= \dot{\alpha}_P - \dot{\gamma}_D = (1 - N_P) \dot{\alpha}_P = \frac{(N_P - 1)(v_E \sin \phi_{LON,E} - v_P \sin \sigma_{LON,P})}{R_{PE}}\end{aligned}\quad (10)$$

Similarly, one may find the rate of the escape angles from Equation (5) by taking their time derivative as follows:

$$\begin{aligned}
\dot{\phi}_{LAT,P} &= \dot{\beta}_D - \dot{\psi}_P = \dot{\beta}_D - N_P \dot{\beta}_P = N_P \frac{(v_E \sin \phi_{LAT,E} - v_P \sin \sigma_{LAT,P})}{R_{PE}} - \frac{(v_P \sin \phi_{LAT,P} - v_D \sin \sigma_{LAT,D})}{R_{DP}} \\
\dot{\phi}_{LON,P} &= \dot{\alpha}_D - \dot{\gamma}_P = \dot{\alpha}_D - N_P \dot{\alpha}_P = N_P \frac{(v_E \sin \phi_{LON,E} - v_P \sin \sigma_{LON,P})}{R_{PE}} - \frac{(v_P \sin \phi_{LON,P} - v_D \sin \sigma_{LON,D})}{R_{DP}} \\
\dot{\phi}_{LAT,E} &= \dot{\beta}_P - \dot{\psi}_E = \dot{\beta}_P - \dot{\psi}_E(t) = \frac{-(v_E \sin \phi_{LAT,E} - v_P \sin \sigma_{LAT,P})}{R_{PE}} - \dot{\psi}_E(t) \\
\dot{\phi}_{LON,E} &= \dot{\alpha}_P - \dot{\gamma}_E = \dot{\alpha}_P - \dot{\gamma}_E(t) = \frac{-(v_E \sin \phi_{LON,E} - v_P \sin \sigma_{LON,P})}{R_{PE}} - \dot{\gamma}_E(t)
\end{aligned} \tag{11}$$

As a result, the fifteen states used to represent the dynamics in Equation (8) are now reduced to ten states in the relative polar space, which is denoted as follows:

$$\begin{pmatrix} \dot{R}_{DP} \\ \dot{R}_{PE} \\ \dot{\sigma}_{LAT,D} \\ \dot{\sigma}_{LON,D} \\ \dot{\sigma}_{LAT,P} \\ \dot{\sigma}_{LON,P} \\ \dot{\phi}_{LAT,P} \\ \dot{\phi}_{LON,P} \\ \dot{\phi}_{LAT,E} \\ \dot{\phi}_{LON,E} \end{pmatrix} = \begin{pmatrix} -v_D (\cos \sigma_{LAT,D} \cos \sigma_{LON,D} + \cos \sigma_{LAT,D} + \cos \sigma_{LON,D} - 1) / 2 \\ + v_P (\cos \phi_{LAT,P} \cos \phi_{LON,P} + \cos \phi_{LAT,P} + \cos \phi_{LON,P} - 1) / 2 \\ -v_P (\cos \sigma_{LAT,P} \cos \sigma_{LON,P} + \cos \sigma_{LAT,P} + \cos \sigma_{LON,P} - 1) / 2 \\ + v_E (\cos \phi_{LAT,E} \cos \phi_{LON,E} + \cos \phi_{LAT,E} + \cos \phi_{LON,E} - 1) / 2 \\ \frac{(N_D - 1)(v_P \sin \phi_{LAT,P} - v_D \sin \sigma_{LAT,D})}{R_{DP}} \\ \frac{(N_D - 1)(v_P \sin \phi_{LON,P} - v_D \sin \sigma_{LON,D})}{R_{DP}} \\ \frac{(N_P - 1)(v_E \sin \phi_{LAT,E} - v_P \sin \sigma_{LAT,P})}{R_{PE}} \\ \frac{(N_P - 1)(v_E \sin \phi_{LON,E} - v_P \sin \sigma_{LON,P})}{R_{PE}} \\ N_P \frac{(v_E \sin \phi_{LAT,E} - v_P \sin \sigma_{LAT,P})}{R_{PE}} - \frac{(v_P \sin \phi_{LAT,P} - v_D \sin \sigma_{LAT,D})}{R_{DP}} \\ N_P \frac{(v_E \sin \phi_{LON,E} - v_P \sin \sigma_{LON,P})}{R_{PE}} - \frac{(v_P \sin \phi_{LON,P} - v_D \sin \sigma_{LON,D})}{R_{DP}} \\ \frac{-(v_E \sin \phi_{LAT,E} - v_P \sin \sigma_{LAT,P})}{R_{PE}} - \dot{\psi}_E(t) \\ \frac{-(v_E \sin \phi_{LON,E} - v_P \sin \sigma_{LON,P})}{R_{PE}} - \dot{\gamma}_E(t) \end{pmatrix} \tag{12}$$

III. Optimal Control

The objective in the active target defense scenario is for the Evader to out-maneuver the Pursuer by using the Defender to his advantage. The objective is to maximize his range from the Pursuer at final time while at the same time minimizing the range of the Defender from the Pursuer at final time. Because the outcome of the scenario considered in this paper is that of successful defense, the final time is defined as the instant where the Defender captures the Pursuer, and therefore $R_{DP}(t_f) = 0$. The objective is therefore:

$$\min_{\mathbf{u}_E} -R_{PE}(t_f) \tag{13}$$

The final time is defined as the instant where the Defender captures the Pursuer, and therefore $R_{DP}(t_f) = 0$.

Using the calculus of variations, one may setup and potentially solve optimal control problems starting with the Hamiltonian. In this case, the objective is Mayer (depends only upon final time) and therefore the Hamiltonian may be obtained by taking the inner-product of the costates with the state-dynamics.

$$\begin{aligned}
\mathcal{H} &= p_{R_{DP}} \dot{R}_{DP} + p_{R_{PE}} \dot{R}_{PE} + p_{\sigma_{LAT,D}} \dot{\sigma}_{LAT,D} + p_{\sigma_{LON,D}} \dot{\sigma}_{LON,D} + p_{\sigma_{LAT,P}} \dot{\sigma}_{LAT,P} + p_{\sigma_{LON,P}} \dot{\sigma}_{LON,P} + \\
&\quad p_{\phi_{LAT,P}} \dot{\phi}_{LAT,P} + p_{\phi_{LON,P}} \dot{\phi}_{LON,P} + p_{\phi_{LAT,E}} \dot{\phi}_{LAT,E} + p_{\phi_{LON,E}} \dot{\phi}_{LON,E}
\end{aligned} \tag{14}$$

Substitution of the dynamics from Equation (12) the Hamiltonian can be written explicitly as a function of states

and costates as follows:

$$\begin{aligned}
\mathcal{H} = & p_{R_{DP}} (-v_D (\cos \sigma_{LAT,D} \cos \sigma_{LON,D} + \cos \sigma_{LAT,D} + \cos \sigma_{LON,D} - 1) / 2 \\
& + v_P (\cos \phi_{LAT,P} \cos \phi_{LON,P} + \cos \phi_{LAT,P} + \cos \phi_{LON,P} - 1) / 2) + \\
& p_{R_{PE}} (-v_P (\cos \sigma_{LAT,P} \cos \sigma_{LON,P} + \cos \sigma_{LAT,P} + \cos \sigma_{LON,P} - 1) / 2 \\
& + v_E (\cos \phi_{LAT,E} \cos \phi_{LON,E} + \cos \phi_{LAT,E} + \cos \phi_{LON,E} - 1) / 2) + \\
& p_{\sigma_{LAT,D}} \left(\frac{(N_D - 1)(v_P \sin \phi_{LAT,P} - v_D \sin \sigma_{LAT,D})}{R_{DP}} \right) + \\
& p_{\sigma_{LON,D}} \left(\frac{(N_D - 1)(v_P \sin \phi_{LON,P} - v_D \sin \sigma_{LON,D})}{R_{DP}} \right) + \\
& p_{\sigma_{LAT,P}} \left(\frac{(N_P - 1)(v_E \sin \phi_{LAT,E} - v_P \sin \sigma_{LAT,P})}{R_{PE}} \right) + \\
& p_{\sigma_{LON,P}} \left(\frac{(N_P - 1)(v_E \sin \phi_{LON,E} - v_P \sin \sigma_{LON,P})}{R_{PE}} \right) + \\
& p_{\phi_{LAT,P}} \left(N_P \frac{(v_E \sin \phi_{LAT,E} - v_P \sin \sigma_{LAT,P})}{R_{PE}} - \frac{(v_P \sin \phi_{LAT,P} - v_D \sin \sigma_{LAT,D})}{R_{DP}} \right) + \\
& p_{\phi_{LON,P}} \left(N_P \frac{(v_E \sin \phi_{LON,E} - v_D \sin \sigma_{LON,P})}{R_{PE}} - \frac{(v_P \sin \phi_{LON,P} - v_D \sin \sigma_{LON,D})}{R_{DP}} \right) + \\
& p_{\phi_{LAT,E}} \left(\frac{-(v_E \sin \phi_{LAT,E} - v_P \sin \sigma_{LAT,P})}{R_{PE}} - \dot{\psi}_E(t) \right) + \\
& p_{\phi_{LON,E}} \left(\frac{-(v_E \sin \phi_{LON,E} - v_P \sin \sigma_{LON,P})}{R_{PE}} - \dot{\gamma}_E(t) \right)
\end{aligned} \tag{15}$$

The costate dynamics are found by taking the following partial derivatives:

$$\begin{aligned}
\dot{p}_{R_{DP}} &= -\frac{\partial \mathcal{H}}{\partial R_{DP}}, \quad \dot{p}_{R_{PE}} = -\frac{\partial \mathcal{H}}{\partial R_{PE}}, \quad \dot{p}_{\sigma_{LAT,D}} = -\frac{\partial \mathcal{H}}{\partial \sigma_{LAT,D}}, \quad \dot{p}_{\sigma_{LON,D}} = -\frac{\partial \mathcal{H}}{\partial \sigma_{LON,D}}, \\
\dot{p}_{\sigma_{LAT,P}} &= -\frac{\partial \mathcal{H}}{\partial \sigma_{LAT,P}}, \quad \dot{p}_{\sigma_{LON,P}} = -\frac{\partial \mathcal{H}}{\partial \sigma_{LON,P}}, \quad \dot{p}_{\phi_{LAT,P}} = -\frac{\partial \mathcal{H}}{\partial \phi_{LAT,P}}, \quad \dot{p}_{\phi_{LON,P}} = -\frac{\partial \mathcal{H}}{\partial \phi_{LON,P}}, \\
\dot{p}_{\phi_{LAT,E}} &= -\frac{\partial \mathcal{H}}{\partial \phi_{LAT,E}}, \quad \dot{p}_{\phi_{LON,E}} = -\frac{\partial \mathcal{H}}{\partial \phi_{LON,E}},
\end{aligned} \tag{16}$$

Define the costate vector, $\mathbf{p} = (p_{R_{DP}} \ p_{R_{PE}} \ p_{\sigma_{LAT,D}} \ p_{\sigma_{LON,D}} \ p_{\sigma_{LAT,P}} \ p_{\sigma_{LON,P}} \ p_{\phi_{LAT,P}} \ p_{\phi_{LON,P}} \ p_{\phi_{LAT,E}} \ p_{\phi_{LON,E}})^T$. Taking the partials from Equation (16), the dynamics of the costates follow the form:

$$\dot{\mathbf{p}} = \mathbf{M} \mathbf{p} \tag{17}$$

where the matrix, $\mathbf{M}$, composed of nonlinear time-varying coefficients is as follows:

$$\mathbf{M} = \begin{bmatrix} 0 & 0 & (N_D - 1)\xi_1 & (N_D - 1)\xi_2 & 0 & 0 & -\xi_1 & -\xi_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & (N_P - 1)\xi_3 & (N_P - 1)\xi_4 & N_P \xi_3 & N_P \xi_4 & -\xi_3 & -\xi_4 \\ \xi_5 & 0 & -(N_D - 1)\xi_6 & 0 & 0 & 0 & \xi_7 & 0 & 0 & 0 \\ \xi_8 & 0 & 0 & -(N_D - 1)\xi_9 & 0 & 0 & 0 & \xi_{10} & 0 & 0 \\ 0 & \xi_{11} & 0 & 0 & -(N_P - 1)\xi_{12} & 0 & -N_P \xi_{12} & 0 & \xi_{12} & 0 \\ 0 & \xi_{13} & 0 & 0 & 0 & -(N_P - 1)\xi_{14} & 0 & -N_P \xi_{14} & 0 & \xi_{14} \\ -\xi_{15} & 0 & (N_D - 1)\xi_{16} & 0 & 0 & 0 & -\xi_{16} & 0 & 0 & 0 \\ -\xi_{17} & 0 & 0 & (N_D - 1)\xi_{18} & 0 & 0 & 0 & -\xi_{18} & 0 & 0 \\ 0 & \xi_{19} & 0 & 0 & (N_P - 1)\xi_{20} & 0 & N_P \xi_{20} & 0 & \xi_{20} & 0 \\ 0 & \xi_{21} & 0 & 0 & 0 & (N_P - 1)\xi_{21} & 0 & N_P \xi_{21} & 0 & \xi_{21} \end{bmatrix} \tag{18}$$

where the time-varying coefficients, $\xi_1 - \xi_{22}$, are as follows:

$$\begin{aligned}
\xi_1 &= \frac{vd \sin \sigma_{LAT,D} - vp \sin \phi_{LAT,P}}{R_{DP}^2} = \frac{\dot{\beta}_D}{R_{DP}} & \xi_2 &= \frac{vd \sin \sigma_{LON,D} - vp \sin \phi_{LON,D}}{R_{DP}^2} = \frac{\dot{\alpha}_D}{R_{DP}} \\
\xi_3 &= \frac{vp \sin \sigma_{LAT,P} - ve \sin \phi_{LAT,E}}{R_{PE}^2} = \frac{\dot{\beta}_P}{R_{PE}} & \xi_4 &= \frac{vp \sin \sigma_{LON,P} - ve \sin \phi_{LON,E}}{R_{PE}^2} = \frac{\dot{\alpha}_P}{R_{PE}} \\
\xi_5 &= v_D (\sin \sigma_{LAT,D} \cos \sigma_{LON,D} + \sin \sigma_{LAT,D})/2 & \xi_6 &= \frac{v_D \cos \sigma_{LAT,D}}{R_{DP}} \\
\xi_7 &= \frac{v_D \cos \sigma_{LAT,D}}{R_{DP}} & \xi_8 &= v_D (\cos \sigma_{LAT,D} \sin \sigma_{LON,D} + \sin \sigma_{LON,D})/2 \\
\xi_9 &= \frac{v_D \cos \sigma_{LON,D}}{R_{DP}} & \xi_{10} &= \frac{v_D \cos \sigma_{LON,D}}{R_{DP}} \\
\xi_{11} &= v_D (\sin \sigma_{LAT,P} \cos \sigma_{LON,P} + \sin \sigma_{LAT,P})/2 & \xi_{12} &= \frac{v_P \cos \sigma_{LAT,P}}{R_{PE}} \\
\xi_{13} &= v_D (\cos \sigma_{LAT,P} \sin \sigma_{LON,P} + \sin \sigma_{LON,P})/2 & \xi_{14} &= \frac{v_P \cos \sigma_{LON,P}}{R_{PE}} \\
\xi_{15} &= v_P (\sin \phi_{LAT,P} \cos \phi_{LON,P} + \sin \phi_{LAT,P})/2 & \xi_{16} &= \frac{v_P \cos \phi_{LAT,P}}{R_{DP}} \\
\xi_{17} &= v_P (\cos \phi_{LAT,P} \sin \phi_{LON,P} + \sin \phi_{LON,P})/2 & \xi_{18} &= \frac{v_P \cos \phi_{LON,P}}{R_{DP}} \\
\xi_{19} &= v_E (\sin \phi_{LAT,E} \cos \phi_{LON,E} + \sin \phi_{LAT,E})/2 & \xi_{20} &= \frac{v_E \cos \phi_{LAT,E}}{R_{PE}} \\
\xi_{21} &= v_E (\cos \phi_{LAT,E} \sin \phi_{LON,E} + \sin \phi_{LON,E})/2 & \xi_{22} &= \frac{v_E \cos \phi_{LON,E}}{R_{PE}}
\end{aligned} \tag{19}$$

In the problem definition, the heading rate and the flight path angle rate of the Evader are constrained. These constrained control requires the use of Pontryagin's Minimum Principle [4, 6] to setup and solve the necessary optimality conditions. The limitation on control are as follows:

$$\begin{aligned}
|\dot{\psi}_E(t)| &\leq \omega_{\psi,max} \quad \forall t \in [t_0, t_f] \\
|\dot{\gamma}_E(t)| &\leq \omega_{\gamma,max} \quad \forall t \in [t_0, t_f]
\end{aligned} \tag{20}$$

Using the definition from Equation (20) and the Hamiltonian defined in Equation (15), the optimal control must satisfy Pontryagin's Minimum Principle [6]:

$$\mathcal{H}(\mathbf{x}^*(t), \mathbf{u}^*(t), \mathbf{p}^*(t), t) \leq \mathcal{H}(\mathbf{x}^*(t), \mathbf{u}(t), \mathbf{p}^*(t), t) \tag{21}$$

The states and costates are defined as follows:

$$\mathbf{x} = \begin{pmatrix} R_{DP} \\ R_{PE} \\ \sigma_{LAT,D} \\ \sigma_{LON,D} \\ \sigma_{LAT,P} \\ \sigma_{LON,P} \\ \phi_{LAT,P} \\ \phi_{LON,P} \\ \phi_{LAT,E} \\ \phi_{LON,E} \end{pmatrix} \quad \text{and} \quad \mathbf{p} = \begin{pmatrix} p_{R_{DP}} \\ p_{R_{PE}} \\ p_{\sigma_{LAT,D}} \\ p_{\sigma_{LON,D}} \\ p_{\sigma_{LAT,P}} \\ p_{\sigma_{LON,P}} \\ p_{\phi_{LAT,P}} \\ p_{\phi_{LON,P}} \\ p_{\phi_{LAT,E}} \\ p_{\phi_{LON,E}} \end{pmatrix} \tag{22}$$

Calculated, the admissible control $\mathbf{u}(t)$ for all $t \in [t_0, t_f]$ can be found to be:

$$\dot{\psi}_E(t)^*(t) = \begin{cases} -\omega_{\psi,max} & p_{LAT,E} > 0 \\ \omega_{\psi,max} & p_{LAT,E} < 0 \end{cases} \quad \dot{\gamma}_E(t)^* = \begin{cases} -\omega_{\gamma,max} & p_{LON,E} > 0 \\ \omega_{\gamma,max} & p_{LON,E} < 0 \end{cases} \tag{23}$$

When either $p_{LAT,E} = 0$ no information about the lateral control, $\dot{\psi}_E(t)$, can be obtained from the Hamiltonian, and a singular trajectory for the optimal heading rate occurs. Similarly, when $p_{LON,E} = 0$, no information about the optimal longitudinal control, $\dot{\gamma}_E$, can be obtained from the Hamiltonian, and a singular trajectory for the optimal flight path angle rate occurs. These singular arcs are problematic for finding the optimal control analytically, and as a result direct methods are required. Considering the boundary conditions of the states and costates, the optimal control problem is formed. According to the objective of this scenario, final time is free and a fixed constraint of $R_{DP} = 0$. The following describes the boundary conditions of the underlying three-agent scenario:

$$\begin{pmatrix} R_{DP}(t_0) \\ R_{PE}(t_0) \\ \sigma_{LAT,D}(t_0) \\ \sigma_{LON,D}(t_0) \\ \sigma_{LAT,P}(t_0) \\ \sigma_{LON,P}(t_0) \\ \phi_{LAT,P}(t_0) \\ \phi_{LON,P}(t_0) \\ \phi_{LAT,E}(t_0) \\ \phi_{LON,E}(t_0) \end{pmatrix} = \begin{pmatrix} R_{DP_0} \\ R_{PE_0} \\ \sigma_{LAT,D_0} \\ \sigma_{LON,D_0} \\ \sigma_{LAT,P_0} \\ \sigma_{LON,P_0} \\ \phi_{LAT,P_0} \\ \phi_{LON,P_0} \\ \phi_{LAT,E_0} \\ \phi_{LON,E_0} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} R_{DP}(t_f) \\ R_{PE}(t_f) \\ \sigma_{LAT,D}(t_f) \\ \sigma_{LON,D}(t_f) \\ \sigma_{LAT,P}(t_f) \\ \sigma_{LON,P}(t_f) \\ \phi_{LAT,P}(t_f) \\ \phi_{LON,P}(t_f) \\ \phi_{LAT,E}(t_f) \\ \phi_{LON,E}(t_f) \end{pmatrix} = \begin{pmatrix} 0 \\ \text{free} \\ \text{free} \\ \text{free} \\ \text{free} \\ \text{free} \\ \text{free} \\ \text{free} \\ \text{free} \\ \text{free} \end{pmatrix} \quad (24)$$

$$\mathcal{H}(\mathbf{x}(t_f), \mathbf{u}^*(t_f), \mathbf{p}^*(t_f), t_f) = 0 \quad (25)$$

Solving the necessary conditions using Equations (12), (17) and (23), provides the general solution to the optimal control problem. Applying the boundary conditions from Equations (24) and (25) provides the so-called 2-point boundary value problem, and consequently the optimal control problem.

IV. Simulation

The three-agent scenario involves the Pursuer, Evader, and Defender in 3-D Cartesian space. The outcome of the scenario greatly varies upon the initial state of the engagement. This includes the location, heading, and flight path angle, and speed of each of the three agents. The aggressiveness by which the Defender and the Pursuer home on their targets is adjusted by the magnitude of their homing gains, N_D and N_P respectively.

Due to the non-linear equations of motion and the large number of states, a numerical tool such as MATLAB can be used to solve the two-point boundary value problem described by the relative polar equations of motion in Equation (12), costate dynamics described in Equations (17) to (19), boundary conditions described in Equations (24) and (25), and constraints described in Equation (20).

Rather than use the single shooting method to pose and solve this optimization problem, the method of even collocation is used. Previous work has described in detail the various advantages of using a collocated optimization routine rather than shooting methods [7, 8, 32]. An evenly spaced collocation method has various advantages over the single shooting method. By transcribing the problem into a static optimization problem, even collocation is less sensitive to initial guesses and is capable of solving optimal control problems in a fraction of the time required by traditional shooting methods. Rather than using an ordinary differential equation solver, the equations of motion are transcribed to a set of equality constraints. These equality constraints are used to form the Karush-Kuhn-Tucker (KKT) conditions. When the equality constraints have been met, then the equations of motion have been satisfied. A flow diagram which describes the Even Collocation Method can be seen in Figure 2 below.

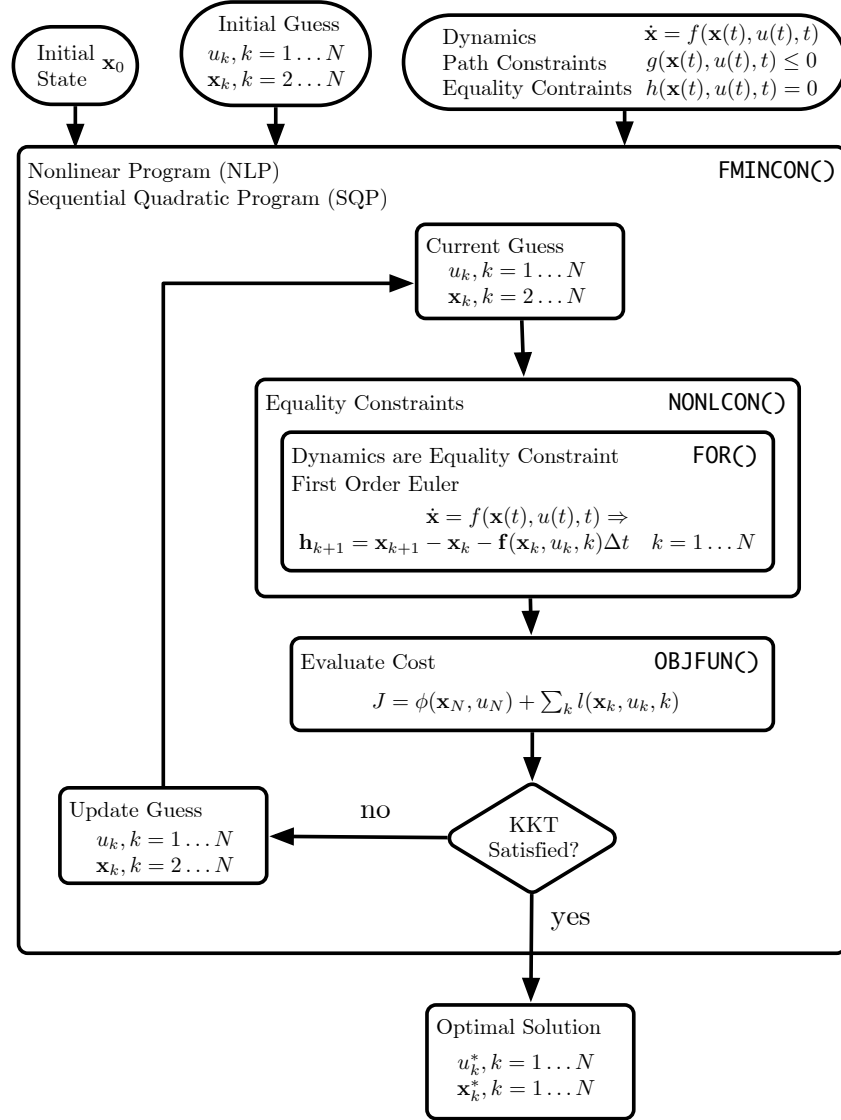


Fig. 2 The Even Collocation Method is a direct transcription method which saves computational effort by transcribing the equations of motion to a set of equality constraints at each of the collocation points [8].

From the figure, the Even Collocation Method (ECM) is described in detail. The optimization routine assumes that there are N evenly spaced collocation points. At each collocation point reside values for the control and state of the dynamic system. In our problem the state-space, $\mathbf{x} \in \mathbb{R}^{10}$, and the control, $\mathbf{u} \in \mathbb{R}^2$. Furthermore, final time is free and therefore the size of the initial guess for N collocation points is $12N + 1$. In order to discretize the equations of motion, the first order Euler method may be used:

$$\mathbf{x}[k + 1] \approx \mathbf{x}[k] + \mathbf{f}(\mathbf{x}[k], \mathbf{u}[k], k)\Delta t \quad (26)$$

Using the discretization, the objective cost functional is simply the Meyer cost contains the final value of the Pursuer-Evader range and the final value of the Defender-Pursuer range:

$$J(\mathbf{x}_k, \mathbf{u}_k, k) = \phi(\mathbf{x}_N, \mathbf{u}_N, N) = R_{DP,N} - R_{PE,N} \quad (27)$$

In code, the equations of motion are written as a set of equality constraints:

$$\begin{aligned}
h_{R_{DP}}[k+1] &= R_{DP}[k+1] - R_{DP}[k] - \dot{R}_{DP}(x[k], u[k], k)\Delta t \\
h_{R_{PE}}[k+1] &= R_{PE}[k+1] - R_{PE}[k] - \dot{R}_{PE}(x[k], u[k], k)\Delta t \\
h_{\sigma_{LAT,D}}[k+1] &= \sigma_{LAT,D}[k+1] - \sigma_{LAT,D}[k] - \dot{\sigma}_{LAT,D}(x[k], u[k], k)\Delta t \\
h_{\sigma_{LON,D}}[k+1] &= \sigma_{LON,D}[k+1] - \sigma_{LON,D}[k] - \dot{\sigma}_{LON,D}(x[k], u[k], k)\Delta t \\
h_{\sigma_{LAT,P}}[k+1] &= \sigma_{LAT,P}[k+1] - \sigma_{LAT,P}[k] - \dot{\sigma}_{LAT,P}(x[k], u[k], k)\Delta t \\
h_{\sigma_{LON,P}}[k+1] &= \sigma_{LON,P}[k+1] - \sigma_{LON,P}[k] - \dot{\sigma}_{LON,P}(x[k], u[k], k)\Delta t \\
h_{\phi_{LAT,P}}[k+1] &= \phi_{LAT,P}[k+1] - \phi_{LAT,P}[k] - \dot{\phi}_{LAT,P}(x[k], u[k], k)\Delta t \\
h_{\phi_{LON,P}}[k+1] &= \phi_{LON,P}[k+1] - \phi_{LON,P}[k] - \dot{\phi}_{LON,P}(x[k], u[k], k)\Delta t \\
h_{\phi_{LAT,E}}[k+1] &= \phi_{LAT,E}[k+1] - \phi_{LAT,E}[k] - \dot{\phi}_{LAT,E}(x[k], u[k], k)\Delta t \\
h_{\phi_{LON,E}}[k+1] &= \phi_{LON,E}[k+1] - \phi_{LON,E}[k] - \dot{\phi}_{LON,E}(x[k], u[k], k)\Delta t
\end{aligned} \tag{28}$$

By appending the collection of equality constraints, the $10N$ equality constraints are formed in the vector,

$$\mathbf{h} = (\mathbf{h}_{R_{DP}}, \mathbf{h}_{R_{PE}}, \mathbf{h}_{\sigma_{LAT,D}}, \mathbf{h}_{\sigma_{LON,D}}, \mathbf{h}_{\sigma_{LAT,P}}, \mathbf{h}_{\sigma_{LON,P}}, \mathbf{h}_{\phi_{LAT,P}}, \mathbf{h}_{\phi_{LON,P}}, \mathbf{h}_{\phi_{LAT,E}}, \mathbf{h}_{\phi_{LON,E}})^T \in \mathbb{R}^{10N}$$

Using the equality constraints in Equation (28), the dynamic optimization problem is transcribed into a static one. Using a nonlinear program (NLP) solver such as FMINCON in MATLAB, one may setup and solve for the $2N$ controls and final time for the optimal control problem.

V. Example

Using the following initial conditions described in Table 1, the optimal control problem may be solved to illustrate the optimal strategy of the Evader which maximizes his range from the Pursuer at final time.

Table 1 Initial Position and Direction of the Three Agents in Cartesian Space

	X-position (x)	Y-Position (y)	Z-Position (z)	Heading (ψ)	Flight Path Angle (γ)
Pursuer	6.0	0.0	5.0	$\pi/2$	0.0
Evader	0.0	11.0	15	$\pi/2$	0.0
Defender	-6.0	0.0	5.0	$\pi/2$	0.0

Using these initial conditions, the relative space representation can be obtained using Equations (3) to (6). Using the smaller state space realization requires fewer variables and numerical solutions are obtained faster as a result of the reduced state space size. Table 2 shows the same initial conditions from Table 1, rather in the relative polar space. Note that the units on the range are a unit of length and the units on the pursuit and escape angles are in radians. For reasons of acceptance, the ranges are a unit of length, but no specific units have been provided.

Table 2 Initial Position and Direction of the Three Agents in Relative Polar Space

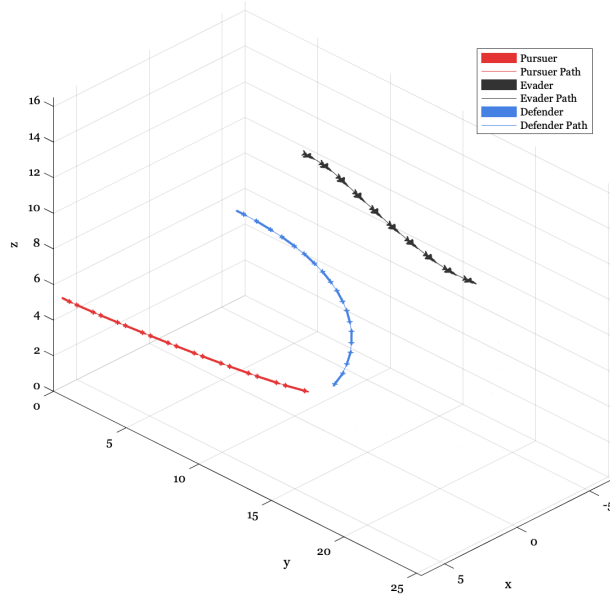
	Range (R)		Pursuit Angles (σ)					Escape Angles (ϕ)		
	R_{DP}	R_{PE}	$\sigma_{LAT,D}$	$\sigma_{LON,D}$	$\sigma_{LAT,P}$	$\sigma_{LON,P}$	$\phi_{LAT,P}$	$\phi_{LON,P}$	$\phi_{LAT,E}$	$\phi_{LON,E}$
Value	12.000	16.0312	-1.5708	0.000	0.4993	0.6736	-1.5708	0.000	0.4993	0.6736

Using the initial conditions described in Table 2, Table 3 shows the constants which are used to completely define the example scenario.

Table 3 Initial Position and Direction of the Three Agents in Relative Polar Space

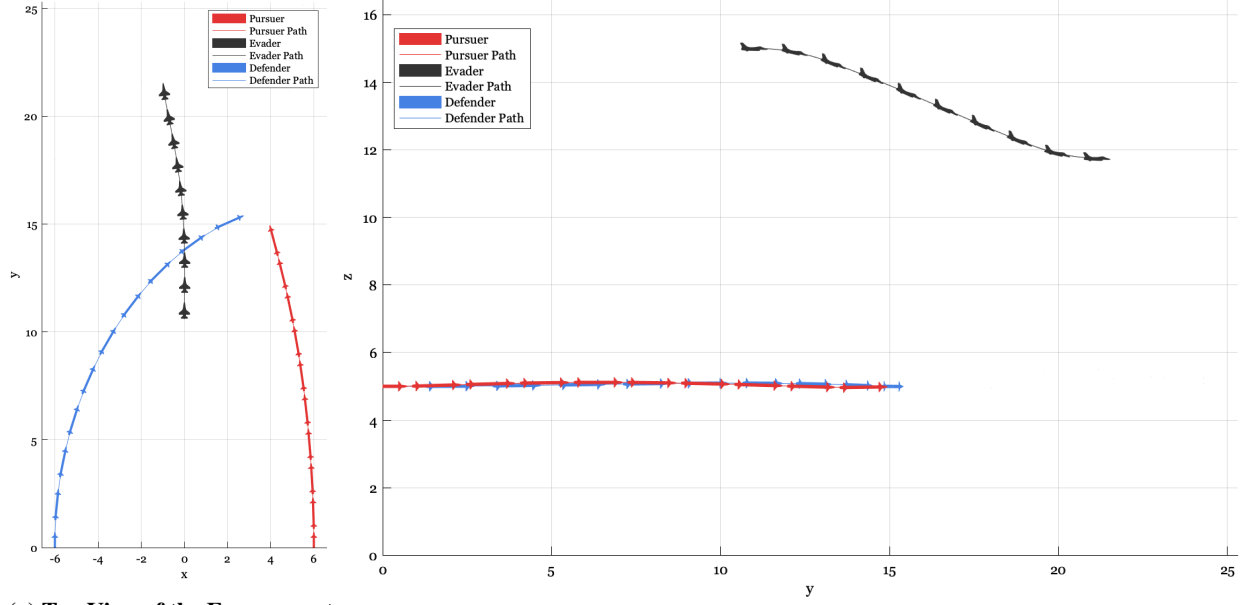
	PN Gain (N)		Velocities (V)			Maximum Turn Rates (ω)	
	N_D	N_P	V_D	V_P	V_E	$\omega_{\psi,max}$	$\omega_{\gamma,max}$
Value	3.00	3.00	1.00	0.80	0.60	0.50	0.10

The optimal control is found by minimizing the objective cost functional in Equation (13) subject to the dynamics in Equation (12) with initial conditions described in Table 2 and terminal boundary constraints shown in Equation (24). Rather than shoot the dynamics forward though time, the even collocation algorithm is used. The optimal control is numerically found by solving the nonlinear program described in Figure 2. By evenly spacing time at set intervals, the dynamics are imposed by the set of equality constraints shown in Equation (28). In order to provide a guess for the nonlinear program solver, FMINCON in MATLAB, the equations of motion in eq. (12) are shot forward though time with zero-input. This guess is fed to FMINCON and a sequential quadratic program is used to find the optimal control. Updates are provided using finite differences, and a single feasible solution (local minimum) is found. Figures 3 and 4 shows the optimal engagement scenario provided by solving for the optimal control directly. In the figure, the black aircraft represents the Evader, the blue vehicle represents the Defender, and the red vehicle represents the Pursuer. Visualization is made possible by free and open-source MATLAB visualization package Flypath3D [33].

**Fig. 3 Three-Agent Engagement Scenario**

Form Figures 3 and 4, one may observe that the Pursuer is unable to capture the Evader and the Defender successfully captures the Pursuer before it can capture the Evader. The maneuver made by the Evader strives at maximizing the separation distance between the Pursuer and Defender at final time and as a result, the observed optimal maneuver is a dive and a slight bank.

Further examination of the Line of Sights and the Optimal control are seen in Figure 5. The line of sight distances between agents is shown in Figure 5a and the optimal controls are shown in Figure 5b. From the line of sight distances, one observes that the maneuver made by the Evader successfully drives R_{DP} to zero while slowing the rate of R_{PE} as much as possible. In order for a successful defense, R_{DP} must go to zero before R_{PE} does. Since $R_{DP}(t_f) = 0$, the final objective cost functional as shown in Equation (13) is in-fact the Pursuer-Evader range at final time, $R_{PE}(t_f)$. Since the velocity of the Evader is less than his Pursuer, the range R_{PE} is monotonically decreasing. This infers that a maneuver by the Evader must be made to leverage his teammate, the Defender, in order to successfully escape capture. As can be observed in Figure 3, such a maneuver is made.

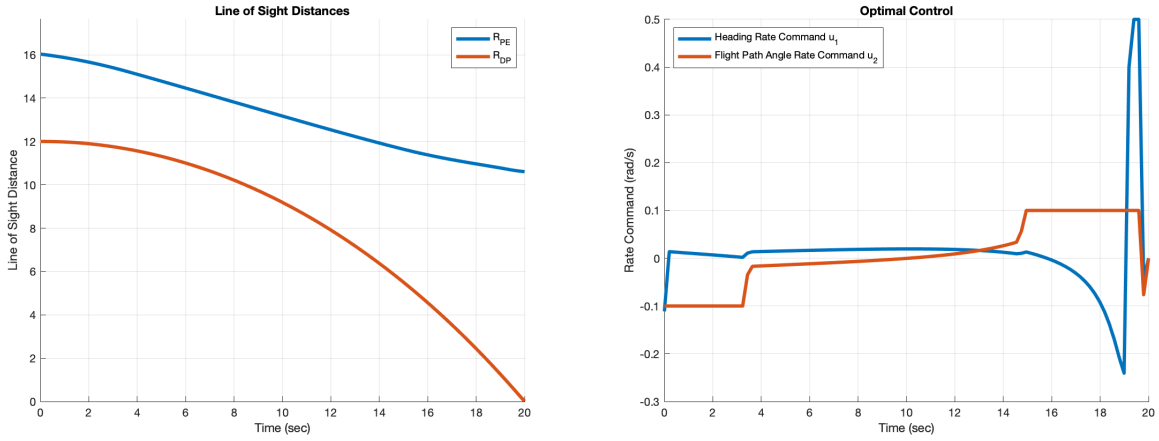


(a) Top View of the Engagement Scenario

(b) Right View of the Engagement Scenario

Fig. 4 Line of Sight and Optimal Control During Engagement Scenario

The optimal maneuver made by the Evader begins with a maximum flight path angle rate as predicted by Equation (23). In the event that the costates in Equation (23) go to zero, the optimal control problem becomes singular and no information is provided about the trajectories of $\dot{\psi}_E(t)^*$ or $\dot{\gamma}_E^*$. This is the case for much of the engagement for the heading rate command $\dot{\psi}_E(t)$. These singular solutions are problematic using indirect methods, but may be handled using the direct methods.



(a) Distances Between Agents

(b) Heading Rate and Flight Path Angle Rate of Evader

Fig. 5 Line of Sight and Optimal Control During Engagement Scenario

VI. Conclusion

In conclusion, a three-agent pursuit-evasion scenario in 3-D consisting of a Pursuer, Defender, and Evader was considered. The objective of the Evader is to out maneuver his Pursuer in order to aid his teammate, the Defender.

Using a proportional navigation law for the both the Pursuer and Defender, an optimal control problem was proposed to find the optimal heading rate and flight path angle rate of the Evader to maximize separation range of the Pursuer when captured by the Defender.

This paper presented the equations of motion in 3-D Cartesian space in Equation (8) and resolved them to a smaller relative polar representation in Equation (12). Using the relative polar formulation, the necessary conditions from Equations (12), (17) and (23), provided the general solution to the optimal control problem. Application of the boundary conditions from Equations (24) and (25) setup the 2-point boundary value problem, and consequently the optimal control problem.

Implementing the equations of motion in MATLAB the optimal control problem was setup and solved using the equal collocation method. An initial guess of a straight forward flight for the Evader was provided using a single shoot of the dynamics. Then, using a nonlinear program solver, the optimal control and states were found and presented using an example scenario.

The successful defense of an Evader from a Faster Pursuer was accomplished by leveraging a teamed Defender. Since the Defender's strategy was one of homing, no information about the Evader was provided to the Defender for successful defense to occur. Moreover, the consideration of 3-D Cartesian space was considered rather than restricting the mathematics to the 2-D plane. This paper concludes that the nonlinear program may be computed and simulation results to the 3-D engagement are found quickly and efficiently using the even collocation method.

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